

also allow bureaucrats to quickly confirm the authenticity of official documents.

Ultra-thin bladder scanner

Researchers at the University of Washington have proposed an ultra-thin laser endoscope, which is like a thin piece of cooked spaghetti, in combination with a software that stitches together the images from the scope's path to create a full, 3D panorama of the bladder interior.

The semi-automated scan could be done by a nurse or technician and the resulting images could be reviewed at a later time, potentially in another city or country. The software makes sure that no part of the organ is missed.

The scope is just 1.5mm wide. The tiny size is possible because of a novel design that swings a single optical fibre back and forth to scan a colour image pixel by pixel. The tip of the device contains a steering mechanism that directs the movement of the scope during the internal exam.

Thumb-sized PC

Imagine a computer that's the size of a USB stick and costs just \$23. Well, your imagination might soon turn into a reality. UK-based game developer David Braben has created a thumb-sized computer called the Raspberry Pi for school children, which is based around a USB stick.

With USB port on one side and HDMI port on the other, it can function as a real PC. This allows a keyboard and a monitor to be connected, respectively. The PC has enough room for a 700MHz ARM11 processor, 128MB RAM and OpenGL graphics with enough power to display 1080p video.

The Raspberry Pi Foundation plans to manufacture and distribute this ultra-low-cost computer for use in teaching computer programming to children.

Foldable display

Researchers at the Samsung Advanced Institute of Technology in South Korea



Computer with a USB-based typewriter

have built the prototype of a seamless foldable active-matrix organic-light-emitting-diode (AMOLED) display that's flexible enough to be folded in half, like a sheet of paper.

The display consists of two AMOLED panels, silicone rubber (a hyperelastic material), a protective glass cover and a module case. It has a very small folding radius of just 1 mm, so one panel lies almost completely on top of the other when the display is folded at a 180-degree angle. The protective glass cover not only prevents the display from scratches but serves as a touch screen as well.

The prototype was tested against 100,000 folding-unfolding cycles to test the junction. The results of the

trial produced no visible crease and the six per cent decrease in brightness at the pivot line was invisible to the human eye.

Type on your PC using a typewriter

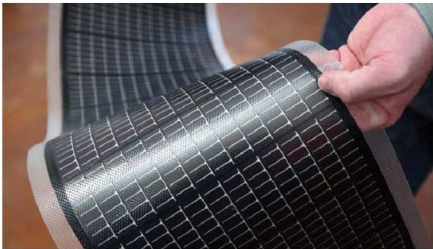
Here is a computer which allows you to use a typewriter instead of a keyboard. Jack Zylkin, a designer in the US, has created a USB typewriter conversion kit which plugs into the USB port on a laptop, monitor or iPad via a sensor board.

The modification is easy to install and involves no messy wiring, and does not change the outward appearance of the typewriter (except for the USB adaptor itself, which is mounted in the rear of the machine).

Solar sheet that captures 90 per cent of available sunlight

A team of researchers led by Patrick Pinhero, associate professor in the chemical engineering department at the University of Missouri, has developed a flexible solar sheet that captures more than 90 per cent of available light.

Called nantenna, the device is a thin, mouldable sheet of small antennae that can harvest the heat from industrial processes and turn it into usable electricity. Since it can absorb infrared radiation, the material has



Solar sheet that captures 90 per cent of sunlight